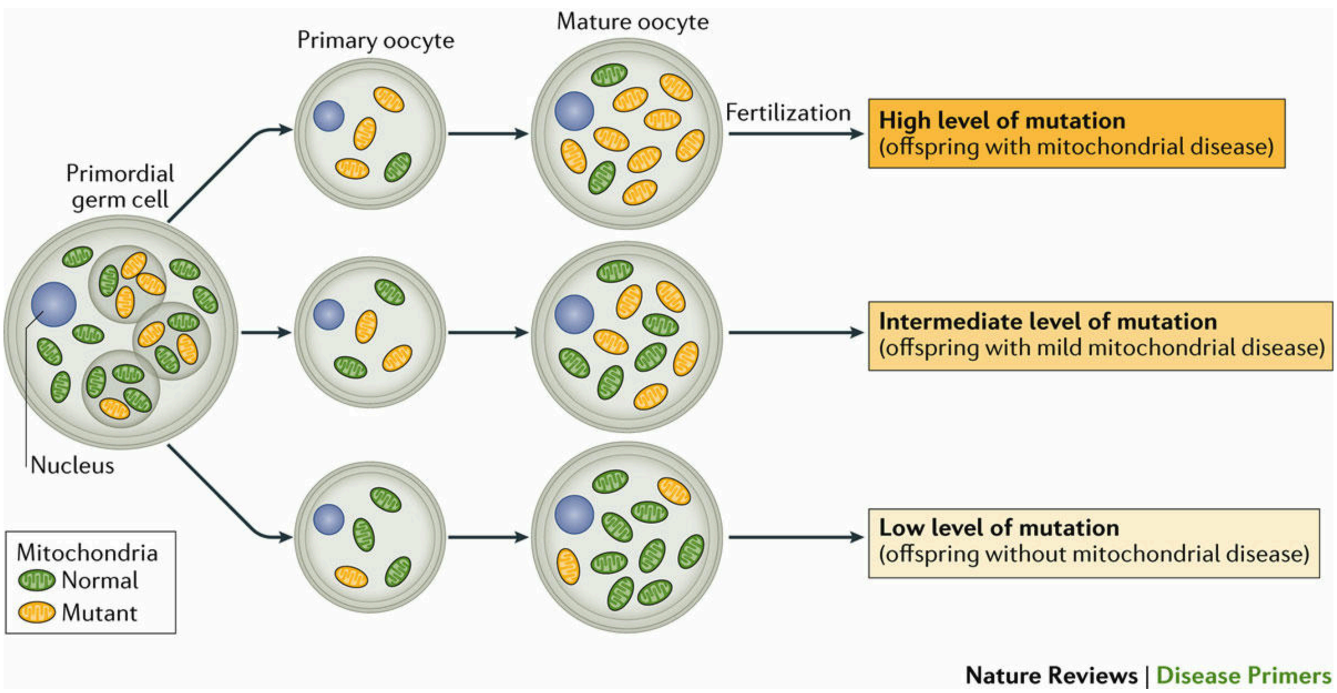
Genetic drift in mitochondrial segregation

Table of contents


```
try:
    import au_molecular_genetics
    print("Already installed")
except ImportError:
    %pip install -q "au_molecular_genetics @ git+https://github.com/au-mbg/molecular_
genetics.git"
```

```
import numpy as np
import matplotlib.pyplot as plt
```

1 Exploring variation from mother to oocytes.



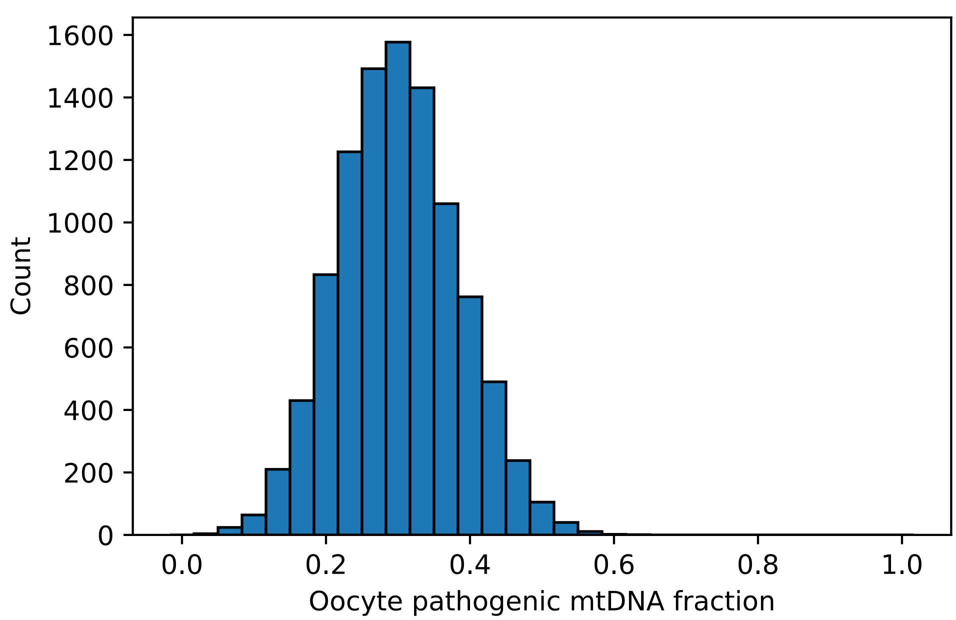
The transmission of heteroplasmic mitochondrial DNA (mtDNA) from mother to offspring is complicated by the genetic bottleneck during development. This bottleneck occurs owing to a profound dilution of mtDNA during the formation of the primordial germ cell (PGC), followed potentially by selective replication of mtDNA genomes. This can lead to profoundly different levels of heteroplasmy in different mature oocytes of women and is an important consideration when counselling mothers with heteroplasmy for pathogenic mtDNA about the risks of having offspring with mitochondrial diseases. The simulation below explores this.

The simulation has two parameters:

- `maternal_heteroplasmy`: This is the fraction of pathogenic mtDNA in the mother's PGC.
- `n_samples`: The number of oocytes sampled, the default of `10000` corresponds to more oocytes than a single woman produces whereas a number like 500 could be considered a snapshot from a single woman.

You may notice that the `sample_oocyte`-function can take an additional parameter `N`, `N` is an effective bottleneck size: the number of mtDNA genomes that effectively found the oocyte population - in humans that number is roughly 30.

```
def sample_oocyte(maternal_heteroplasmy, n_samples=1, N=30):
    oocyte_sample = np.random.binomial(n=N, p=maternal_heteroplasmy, size=n_samples)
    p_oocyte = oocyte_sample / N
    return oocyte_sample, p_oocyte


## Parameters
maternal_heteroplasmy = 0.3 # Maternal heteroplasmy.
n_samples = 10000 # Number of oocytes sampled.

## Sampling
sample, p_ooc = sample_oocyte(maternal_heteroplasmy, n_samples)

## Plot
fig, ax = plt.subplots()
N = 30
bins = np.linspace(-0.5/N, 1 + 0.5/N, N + 2)
ax.hist(p_ooc, bins=bins, edgecolor='black')
ax.set_xlabel('Oocyte pathogenic mtDNA fraction')
ax.set_ylabel('Count')
plt.show()
```



Exercise 1

The number of oocytes produced by a woman during her lifespan is roughly 500. Run the simulation multiple times with `n_samples = 500` and consider the variation in oocyte outcomes for mothers with the same `maternal_heteroplasmy`. Does the *true* distribution change, or just our estimate of it?

If women produced fewer oocytes, say 100, what changes about what you observe from one woman if she produces 100 oocytes instead of 500?

Exercise 2

Vary `maternal_heteroplasmy` through `0.1`, `0.3` and `0.6`. How does the center of the distribution change? Does the width of the distribution change significantly?

Exercise 3

Two women have the same average pathogenic mtDNA fraction. One gives birth to a set of children with very different outcomes, the other is giving birth to a set of children with similar outcomes. Explain how that can happen?

2 Disease model

A simple model of the risk of disease development in a child, based on the fraction of pathogenic mtDNA in the oocyte is a threshold model. That is

- **Case 1:** Low disease allele frequency ($f < t_1$) $\rightarrow$ Healthy child
- **Case 2:** Intermediate disease allele frequency ($t_1 \leq f < t_2$) $\rightarrow$ Mildly affected child
- **Case 3:** High allele frequency in the oocyte ($f \geq t_2$) $\rightarrow$ Severely affected child

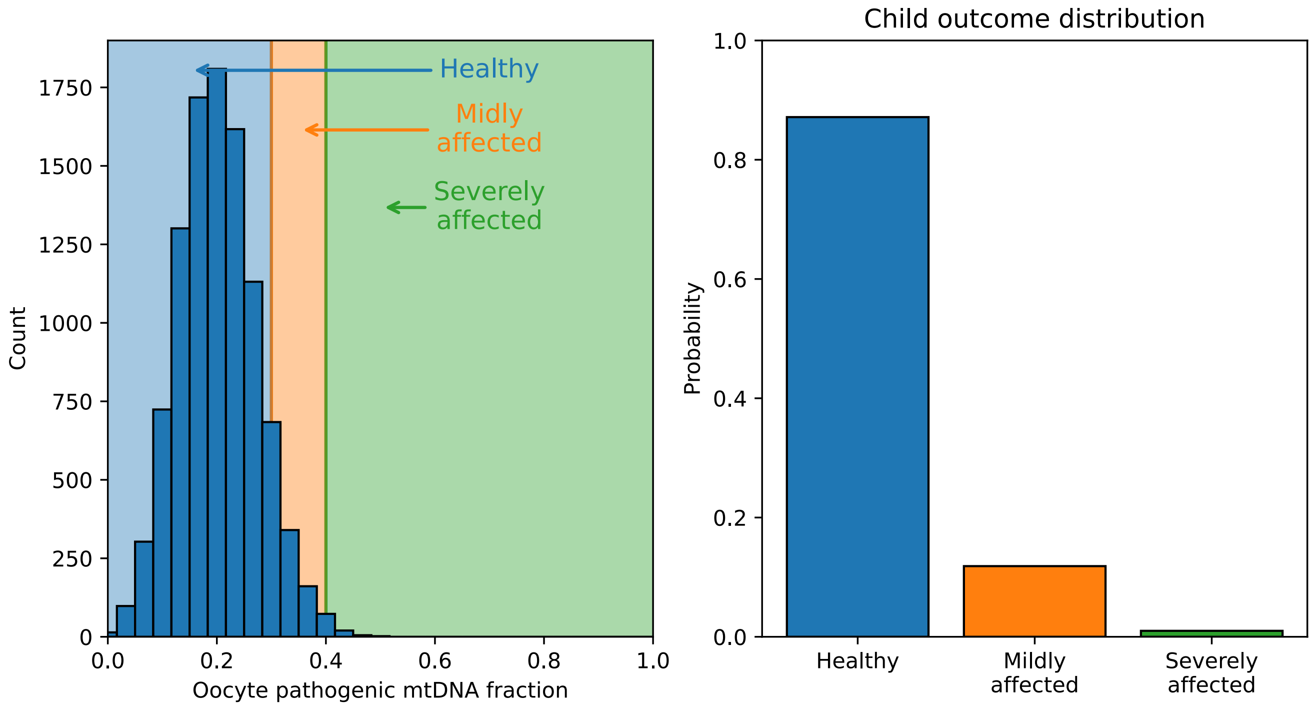
The cell below implements such a model, where you can control the parameters of oocyte sampling and the threshold model.

```
def threshold_model(p, t1, t2):
    state = np.zeros_like(p)        # Case 1
    state[(p >= t1) & (p < t2)] = 1 # Case 2
    state[p >= t2] = 2              # Case 3
    return state

def model_plot(p_ooc, child_state, t1, t2, bins, n_samples):

    states = {0: 'Healthy', 1: 'Mildly\naffected', 2: 'Severely\naffected'}

    ## Plotting
    fig, axes = plt.subplots(1, 2, figsize=(10, 5))

    ## Left plot of oocyte distributions
    ax = axes[0]
    ax.hist(p_ooc, bins=bins, edgecolor='black') # Plots the distribution

    ## Adds lines and shading
    ax.axvline(t1, color='C1', zorder=-1)
    ax.axvline(t2, color='C2', zorder=-1)
    ax.axvspan(0, t1, color='C0', alpha=0.4, zorder=-1)
```

```
    ax.axvspan(t1, t2, color='C1', alpha=0.4, zorder=-1)
    ax.axvspan(t2, 1, color='C2', alpha=0.4, zorder=-1)

    # Annotation arrows.
    ax.annotate('Healthy', (t1/2, 0.95), (0.7, 0.95), xycoords=(ax.transAxes,
ax.transAxes), arrowprops=dict(arrowstyle='->', color='C0', lw=1.5), va='center',
ha='center', color='C0', fontsize=12)
    ax.annotate('Midly\naffected', ((t1+t2)/2, 0.85), (0.7, 0.85), xycoords=(ax.transAxes,
ax.transAxes), arrowprops=dict(arrowstyle='->', color='C1', lw=1.5), va='center',
ha='center', color='C1', fontsize=12)
    ax.annotate('Severely\naffected', (t2+0.1, 0.72), (t2+0.3, 0.72),
xycoords=(ax.transAxes, ax.transAxes), arrowprops=dict(arrowstyle='->', color='C2',
lw=1.5), va='center', ha='center', color='C2', fontsize=12)

    ## Labels and limits
    ax.set_xlabel('Oocyte pathogenic mtDNA fraction')
    ax.set_ylabel('Count')
    ax.set_xlim([0, 1])

    ## Right plot of child outcomes.
    ax = axes[1]
    for state, desc in states.items():
        # Makes a bar for each category by counting the number in each state.
        ax.bar(state, np.sum(child_state == state)/n_samples, width=0.8, edgecolor='black')

    ax.set_xticks(list(states.keys()))
    ax.set_xticklabels(list(states.values()))
    ax.set_ylabel('Probability')
    ax.set_title('Child outcome distribution')
    ax.set_ylim([0, 1])
    plt.show()
```

The next cell produces a plot using the model with the given parameters

```
## Parameters
maternal_heteroplasmy = 0.2
n_samples = 10000

t1 = 0.3 # Disease threshold
t2 = 0.4 # Severe threshold

## Run the simulation
sample, p_ooc = sample_oocyte(maternal_heteroplasmy, n_samples)
child_state = threshold_model(p_ooc, t1, t2)
model_plot(p_ooc, child_state, t1, t2, bins, n_samples)
```



Exercise 4

With `maternal_heteroplasmy = 0.2`, `t1 = 0.3` and `t2 = 0.4`, the mother's fraction of pathogenic mtDNA (`maternal_heteroplasmy`) is below the disease threshold (the mother is healthy). Why do mildly or severely affected children still appear in the simulation?

Exercise 5

Based on the simulations can you explain how a disease caused by pathogenic mtDNA might skip a generation? That is for example a woman that has several healthy children but some of her grandchildren are diseased.